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An assistive haptic-based obstacle avoidance system for individuals with profound visual impairment

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ABSTRACT

This paper presents a lightweight and low-cost wireless electronic system that helps users with profound visual impairment avoid obstacles in an unknown landscape. The system comprises two parts: (1) the 'sensor module' which is laser-based and can be worn on the belt, and (2) two 'haptic modules' which can be worn on each wrist with its vibrators directly placed on the skin. These two parts communicate wirelessly, which enables the user to comfortably and discretely wear the system. The sensor module creates a representation matrix of the surrounding objects. As an object gets closer, a higher pulse-width-modulation (PWM) duty cycle (i.e., a higher vibratory power) is delivered by the corresponding vibrator(s), indicating the object's direction and proximity. This obstacle avoidance system was tested on seven individuals between 50 and 78 years of age (mean = 63.5, SD = 9.3 years) with profound vision impairment. The participants were instructed to walk through two path configurations: one with and another without the proposed assistive system. According to our results, significantly ($p < 0.05$) shorter time (Median = 64s (IQR 32–81) versus Median = 95s (IQR 51–134)) was needed and slightly fewer collisions occurred once they used this assistive system.

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Introduction

According to the World Health Organization (WHO, 2023), over 2.2 billion people have near or distance vision impairment worldwide, and most people with profound vision impairment or blindness are over 50 years old. It has also been reported that profound vision impairment severely affects people's personal lives and has an enormous economic impact on society (WHO, 2023). One of the challenges faced by people with profound visual impairment in their daily lives is safe and independent mobility in an unfamiliar (or less familiar) environment, which can have a negative impact on their quality of life (Bhowmick & Hazarika, 2017; Brouwer et al., 2008; Kim & Cho, 2013).

In recent decades, many assistive devices have been developed to address this mobility problem for visually impaired people (Bhowmick & Hazarika, 2017; Lavric et al., 2024; Muhsin et al., 2024; Panwar et al., 2024; Roentgen et al., 2009). They have utilized cutting-edge technologies such as inertial measurement units (IMUs), light detection and ranging (LiDAR) sensors, the global positioning system (GPS), image processing, artificial intelligence, and visible light communications in their designs. Despite great efforts to introduce advanced technology, the white cane has been the most dominant aid (Levesque, 2005; Sato et al., 2019). However, white canes have some limitations. For instance, they rely on the hand's constant movement, which might be tiresome over a long period, especially for elderly people. In addition, owing to its short range, the white cane only helps find obstacles at a close distance of approximately one meter. Moreover, it may miss objects that do not have a footprint on the ground surface. To further assist visually impaired people with mobility, several electronic travel aids (ETAs) based

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on either ultrasonic or optical sensor systems have been designed. UltraCane (Sound Foresight Technology Ltd, [n.d.](#)) and MiniGuide (GDP Research, [n.d.](#)) are commercial products in which ultrasonic sensors measure the distance to nearby objects with a maximum range of 4 m and 8 m, respectively. A user evaluation of these two electronic mobility aids was conducted by Roentgen et al. (2012a). They concluded that the walking speed generally declined when using these aids compared with the white cane, but significantly fewer collisions occurred.

Based on the literature, despite some effectiveness, ultrasonic assistive devices are, to some extent, limited concerning obstacle detection and the differentiation of targets, and therefore, not adequately reliable in all circumstances (Bhatlawande et al., 2014; Cheng, 2016; Shoval et al., 2003). A limitation of any ultrasonic system is that it cannot provide detailed information about the surroundings because of the small distance that its signal can travel and its inherent latency (Jain & Patel, 2023). In contrast, a focused optical beam such as a laser has a larger range and higher resolution, enabling the identification of precise directions with higher accuracy (Röijezon et al., 2019). The reason for this distinction is the fundamental difference between the two wave types. Ultrasonic sensors are based on acoustic (mechanical) waves that diffuse extensively and attenuate rapidly as they propagate (Carlson et al., 2008) whereas optical waves are electromagnetic waves that can travel much longer distances with only minimal attenuation and divergence (Khodadad et al., 2023). Teletact (Farcy & Bellik, 2002) is an example of commercial ETA in which a laser diode and an array of photoelectrodes are used to emit and detect the laser light respectively.

LiDAR (Light Detection and Ranging) is a technology to measure long-range distances that was initially developed for autonomous vehicles. Recently, several research on navigation systems for visually impaired people have used LiDAR in their designed hands-free systems (Chai & Lau, 2020; Liu et al., 2021; Maryono et al., 2021; Setiadi et al., 2020). Although the LiDAR-based systems at the research level can provide high accuracy in detecting obstacles and distances, they are very heavy-weight and costly because of their dependency on large high-speed computing systems like graphical processing units (GPUs). As a result, these LiDAR-based systems are currently considered infeasible in real-time situations as user-friendly wearable devices (Muhsin et al., 2024). In addition, poor efficiency for reflective surfaces or clear glasses was reported as another limitation of LiDAR-based navigation systems (Jain & Patel, 2023).

When transforming spatial information for visually impaired persons, haptic or touch-based feedback has an advantage over auditory feedback (Cheng, 2016; Innala Ahlmark, 2016; Levesque, 2005). This is because feedback in the form of an audio signal might be too obtrusive or block other important environmental sounds for visually impaired users. Haptic feedback in different forms and on different body locations has been used in research prototypes to assist blind navigation. Altini et al. (2011) utilized haptic stimuli to the user's chest and shoulders. Adame et al. (2013) compared different haptic devices in terms of vibration temporization and intensities, for guiding blind individuals. Bhatlawande et al. (2014) designed an ultrasonic assistive system in which the distance to the obstacle was proportional to the vibration magnitude on a customized bracelet. A haptic-based guidance system was proposed by Scheggi et al. (2014). Their system, however, depended on a remote operator to provide directional information by activating vibrotactile stimulations on the user's two bracelets. Innala Ahlmark (2016) and Röijezon et al. (2019) developed a haptic navigation aid using a prototype named virtual white cane. The virtual white cane was limited to an autonomous wheelchair equipped with a laser system able to determine the distance to every object up to 20 m in a horizontal plane. A recent study designed an ETA using a backpack-based haptic belt, a ZED stereo camera mounted in front of a commercial backpack, and a laptop inside the backpack to enable reconstruction of the depth field from acquired images (Boldini et al., 2021). The haptic feedback was via an array of macro fiber composites (MFC actuators) connected to a computer vision system through a channel of wires. Their study claimed to provide real-time object detection utilizing a neural network algorithm. However, the practical usability of their proposed system in daily life is questionable due to the size and weight of equipment that users need to carry.

Objectives

This study aims to introduce a low-cost, discrete, and light-weight assistive device as a complement to the white cane and to investigate the potential benefits of the designed system for individuals with profound visual impairment. The specific objectives of our design are (1) to deploy laser sensors to detect multiple objects simultaneously, (2) to provide a clear and intuitive haptic feedback signal to multiple vibrators placed on the user's wrist for coding the surrounding objects, (3) to provide a fully wireless connection between the sensor and haptic modules to offer comfort to the user when they wear the system, and (4) to use cost-effective and light-weight electronic components and sensors in order to limit the weight of modules to 200 g and the production cost of the system to under 50 USD.

Justification of need

According to a recent review study, despite many research prototypes, there is currently no ideal user-friendly assistive tool for visually impaired people featuring low-cost and efficient technologies (Muhsin et al., 2024). In addition, according to their review, neither the researchers nor the assistive device developers were successful in identifying the human aspects of blind people's needs such as usability, learnability, and time to user adaption. These limitations, among others, make most of the designed prototypes to be still far from being practical in the everyday life of blind individuals. Further, as visually impaired people usually do not have a good financial situation in society, special attention must be paid to choosing electrical equipment and technology for designing assistive tools to be economically affordable [a good example is the system suggested by Dragne et al. (2022)].

In this study, we introduce a haptic-based assistive device that consists of laser sensors instead of ultrasonic sensors to provide relatively higher accuracy. Our design enables the device to simultaneously detect multiple objects and wirelessly code them using multiple vibrations to provide detailed and intuitive real-time information about the surroundings. To the best of our knowledge, no published study has presented an assistive system that has the following features together: wireless, light-weight, wearable, haptic-based, real-time, using simple laser sensors, and that is also economically affordable for visually impaired users [see Panwar et al. (2024) and Muhsin et al. (2024) for a review]. Although many research prototypes possess a number of these attributes, owning all these features together makes our design rather unique in this field.

Design

Sensor module

The 'sensor module' consists of two laser breakout boards (based on VL53L5CX sensor components from STMicroelectronics) which are placed at 90° angle relative to each other. These far-range sensors measure directions and distances to all objects in their field of view (FoV) based on an infrared transmitter and receiver using diffractive optical elements. Each sensor can simultaneously measure the distance to multiple objects within a range of 2 cm and 400 cm and categorize their corresponding directions by 8 × 8 separate zones in its output.

The 'sensor module' also comprises a microcontroller (ESP32 based on Xtensa processor) with an integrated wireless unit and a rechargeable Lithium battery with 750 mAh capacity. The algorithm on the microcontroller sends a 'representation matrix' of 8 × 8 (corresponding to the two-dimensional FoV of each laser) to a 'haptic module'. The 'sensor module' can be worn by the user on their clothes, for example, on the outer side of the belt, as shown in Figure 1. The microcontrollers were programmed using ESP-IDF via Visual Studio Code in C.

Haptic module

There are two 'haptic modules', one placed on the user's right wrist and the other on their left wrist using a hand band for each. Six vibrator actuators were configured in a 2 × 3 array (as shown in Figure 1) for each 'haptic module'. The horizontal distance between the vibrators is 2 cm within a row.



Figure 1. (a) A participant wearing the presented assistive system. The sensor module is placed on the outside of the belt, and the two haptic modules are placed one on the right and another on the left wrist. (b) A closer look at a haptic module and its vibrators.

The vertical rows are 4 cm apart from each other. This array configuration enables the coding of both the direction and the distance to the objects in a manner that can be naturally understood by humans. The vibrator actuators are DC motors of type eccentric rotating mass (ERM) produced by Seeed studio under the brand name 'mini vibration motor, 316040001'. When connected to a 3.7 VDC supply voltage, this actuator generates a vibration at an intrinsic fundamental frequency of 115 Hz on its surface. The drive circuit implemented in this system consists of six separate N-channel MOSFETs with associated resistors and protection diodes.

Each 'haptic module' also comprises a 750 mAh rechargeable Lithium battery and a microcontroller (ESP32 based on Xtensa processor) with an integrated wireless unit to receive the 'representation matrix' from the 'sensor module' over a wireless radio frequency of 2.4 GHz. Based on the content of the 'representation matrix', the algorithm implemented on the microcontroller decides to vibrate a certain vibrator with a certain pulse-width modulation (PWM) duty cycle, so that the 8×8 elements in the 'representation matrix' corresponding to the FoV of the sensor are transformed to the 2×3 elements corresponding to the vibrators in the 'haptic module'. For instance, if there is an object on the right side of the user, it is detected by the laser sensor located towards the right, and then coded through the vibrator(s) on the right wrist depending on how far the object is located from the middle.

The PWM technique is capable of delivering a higher and steadier power than the simple magnitude or frequency coding and thus is predominantly used in control engineering to efficiently transmit power from electrical systems to mechanical loads (Butterfield & Szymanski, 2018; Sun, 2012). The constant frequency of the PWM signal created on the 'haptic module' was set to 89 Hz. This pulse frequency is applied when the user is within a certain distance from the surrounding objects. The duty cycle of the PWM signal is a function of the distance to the surrounding object (according to Eq. (1.a)) and can be varied in the range of 0 to approximately 98.4% (corresponding to $d = 2$ cm which is the shortest distance coded by the laser module). The closer the detected object, the higher the duty cycle (α) of the PWM signal. Higher duty cycles deliver higher vibratory power to the skin of the user. Furthermore, although the fundamental frequency is kept constant, the harmonic frequency components generated by the PWM do vary as the pulse width (related to α) varies as a function of distance.

$$\alpha = \text{duty cycle} = \begin{cases} \frac{n-d \times 2}{n}, & \alpha > 0 \\ 0, & \alpha \leq 0 \end{cases} \quad (1.a)$$

$$P_{electric} = \frac{(V_s \times \alpha)^2}{R_{eq}}, P_{mech} \cong 0.9 \times P_{electric} \quad (1.b)$$

In Eq. (1.a), 'd' stands for the distance value in cm and the 'n' parameter is one unit larger than twice the maximum distance that the system reacts to. In this study, the 'n' parameter was set to 255, meaning that the maximum distance that the user cares about when walking inside a building is approximately 127 cm. The electrical power produced by the PWM wave ($P_{electric}$) as a function of the duty cycle (α) is given by Eq. (1.b) where V_s is the supply voltage of the vibrator and is constant (3.7 V). R_{eq} is the equivalent resistance of the electrical circuitry which was measured 42 Ω under operation. The electrical circuitry of the vibrator also includes an inductive impedance which can be discarded given the low-frequency nature of the presented system. Given the high efficiency of electrical actuators and the tight mechanical coupling between the vibrator's body and the skin surface, it is reasonable to assume that about 90% of $P_{electric}$ is converted to mechanical power (P_{mech}) which is then delivered to the user's skin.

The communication between the 'sensor module' and the two 'haptic modules' is wireless using the built-in wireless unit on the ESP32 board. This provides substantial freedom to the user by allowing them to conveniently wear the modules.

Precision and sensitivity parameters

The precision of the system was calculated using Eqs. (2) and (3) as follows. There are two laser sensors in the 'sensor module'. Each laser sensor covers a zone of 8×8 elements. Considering the 65° diagonal FoV (45° in the horizontal axis, as depicted in Figure 2a) for each sensor and the 90° relative angle to each other, together they cover approximately 90° in the azimuth plane, as depicted in Figure 2b. Therefore, each element of the 8×16 combined FoV of the two sensors, that is Δs in Eq. (3), was approximately 6° in the azimuth plane. Accordingly, for objects that are for instance 1 m away from the sensors, each element in the two-dimensional FoV or the 'representation matrix' corresponds to approximately $10 \times 10 \text{ cm}^2$ ($\Delta b = 10 \text{ cm}$ in Eq. (2)) of the space in front of the sensors. This leads to encoding the objects with high precision.

According to WHO's guidelines, a standard white cane should be long enough to reach the breast-bone when vertically held. Therefore, for an average person with a height of 1.75 m, the length of the white cane should be around 1.5 m and the 'sensor module' is worn at the height of about 1 m. The system's FoV is 65° in the elevation plane (Figure 2a). As a result, at a ground distance of 1.5 m, an object with a minimum height of 3.7 cm from the floor will be detected by the system.

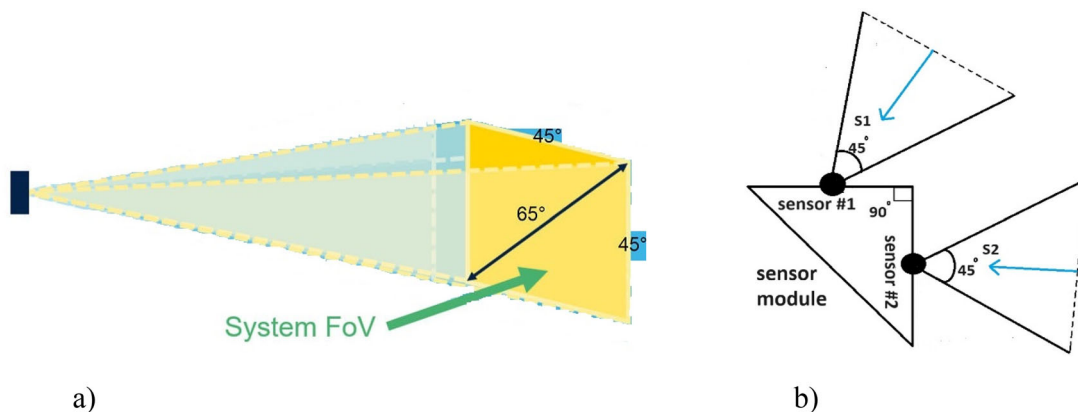


Figure 2. (a) A description of the laser's Field of View (FoV) according to the datasheet (STMicroelectronics, 2023). (b) A schematic of the combined horizontal FoV of the two laser sensors (S1 and S2) in the azimuth plane.

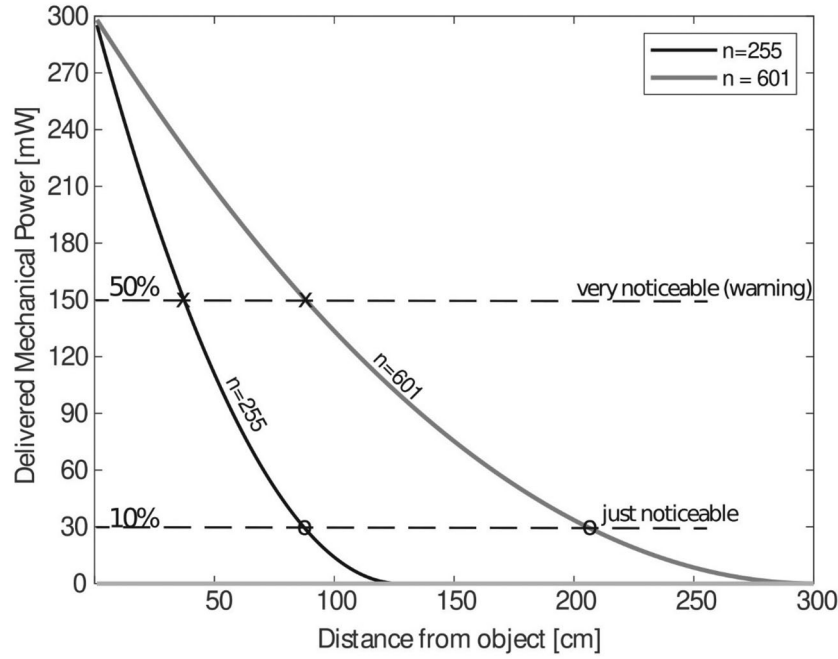


Figure 3. The power delivered by the haptic modules as a function of distance for two different ‘n’ settings.

$$\text{precision } \Delta b = 2 \times 100 \times \tan \frac{\Delta s}{2} [\text{cm}] \quad (2)$$

where

$$\Delta s = \frac{45^\circ + 45^\circ}{16} \times \frac{\pi}{180} [\text{rad}] \quad (3)$$

The sensitivity of the system is defined as the range of the distance in which the surrounding objects are coded by ‘haptic modules’. The minimum distance was 2 cm, and the maximum distance was determined by choosing the ‘n’ parameter in Eq. (1.a), depending on the maximum distance that the user cares about. For instance, the maximum distance could be set to 3 m for outdoor mobility if $n = 601$ was chosen in Eq. (1.a). This means that the system does not code (i.e. duty cycle is zero) any object further away than 3 m. The upper limit for ‘n’ is 801 corresponding to 400 cm i.e., the maximum range coded by the laser.

Figure 3 shows the mechanical power delivered to the user’s skin (P_{mech}) by the ‘haptic module’ as a function of distance for two ‘n’ settings. According to Eq. (1.b), the maximum of P_{mech} corresponds to the duty cycle of 100% ($\alpha = 1$) and is about 300 mW. The solid line shows the power-distance relation for $n = 255$. Accordingly, the system codes distances up to 127 cm, and farther objects are ignored. However, only at about $d = 90$ cm (marked by a circle), the corresponding haptic vibrator starts to deliver 30 mW (i.e. 10% of its maximum power). We observed that this power level is perceived as a ‘just noticeable’ level. From this point, as the distance to objects reduces, the power grows exponentially. At $d = 39$ cm (marked by a cross), the delivered power is at 150 mW (i.e. 50% of its maximum) which is very noticeable and can be regarded as a ‘warning’ signal.

The grey curve in Figure 3 illustrates the power-distance relation for $n = 601$. Accordingly, objects that are farther than 300 cm are ignored. The ‘just noticeable’ level occurs at $d = 205$ cm and the ‘warning’ level is reached at $d = 88$ cm. The comparison between the solid curve and the grey curve suggests that when ‘n’ is chosen 255, the operational range of the system becomes much shorter which appears like an appropriate choice for indoor settings. In contrast, setting ‘n’ to 601 drastically widens the operational range and shifts it to longer distances. This suggests that $n = 601$ could be a good choice for outdoor conditions.

Methods

Test subjects

The designed assistive system was tested on visually impaired people. The test subjects were recruited with the help of a visual-impaired association (Synskadades Riksförbund) in Västerbotten, Sweden. All participants signed a written consent form to participate in the study; and to consent that their data would be anonymously used in scientific analyses and publications. Participants with profound blindness were able to use various digital assistant systems to read out and understand textual information. The presented work is a non-interventional non-invasive study in which seven test subjects tested a consumer product prototype designed to improve their mobility capabilities. There is no risk to the individual or harm to the environment associated with this study. The procedure was done according to the national ethical regulations by the Swedish Ethical Review Authority (Etikprövningsmyndigheten), 2023. No formal ethics review of the study was required based on the institutional policies of the Department of Applied Physics and Electronics at Umeå University.

The inclusion criteria were an age of less than 80 years and a medical diagnosis of profound visual impairment. This age range was chosen because all people with visual impairment registered by the association were over 50 years old. Our exclusion criteria were any cognitive impairment or other sensorimotor impairments that precluded the user from taking the tests. Candidates who fulfilled our criteria received a written invitation letter from the visual-impaired association. Seven individuals aged between 50 and 78 years (mean = 63.5, SD = 9.3) volunteered to participate in this study.

Test setup

The participants were asked to walk through two different path configurations: one with and another without using the assistive system. Participants with a prescribed white cane were allowed to use it during the test. In this case, which was true for six of the seven participants, the participant had a white cane as their baseline on both paths; therefore, the system was tested as a complement to the white cane. Each path was approximately 20 m long and consisted of chairs and tables. Figure 4 illustrates a schematic of the two path configurations. The order of the two tests and the path configurations were alternated among the participants to avoid bias by the system or the individual's experience. The sensitivity of the assistive system was equal for all participants throughout the tests by setting the maximum coded distance to 127 cm (i.e. $n = 255$ in Eq. (1)). Each participant began a test from a start point, was allowed to go either straight ahead or to the sides (not backward), and was informed when they arrived at the destination (stop point). Then, while they were waiting, the test instructor changed the path configuration for the second test to start from the previous stop point. The participants were not provided with any other assistance during the test. In each test, the number of collisions and time taken by the participant to pass through the path were recorded.

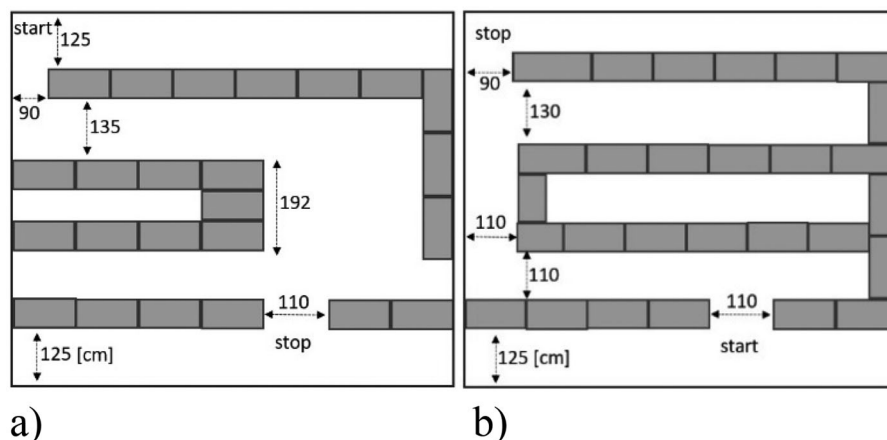
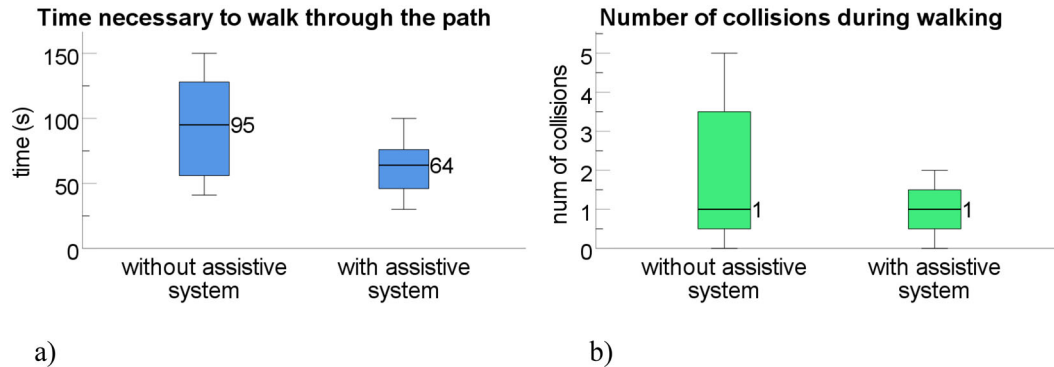


Figure 4. Schematics of the two path configurations (a) and (b).

Table 1. The questionnaire used in this study (it was originally in Swedish but here it is translated into English).

Questions	Answers
1. Did the assistive system make it easier to navigate the room?	No, not at all (1)/Yes, much simpler (10): 1 2 3 4 5 6 7 8 9 10
2. Did the vibration signals feel intuitive?	No, not at all (1)/Yes, really (10): 1 2 3 4 5 6 7 8 9 10
3. Did you feel a sense of trust in the system?	No, no confidence (1)/Yes, big confidence (10): 1 2 3 4 5 6 7 8 9 10
4. Could you distinguish vibrations on the right wrist from the vibrations on the left wrist?	No, not at all (1)/Yes, without a doubt (10): 1 2 3 4 5 6 7 8 9 10
5. Could you determine whether the assistive system vibrated more to the right or left on the same wrist?	No, not at all (1)/Yes, without a doubt (10): 1 2 3 4 5 6 7 8 9 10
6. What potential do you see in a smart haptic-based navigation system?	No potential (1)/Very high potential (10): 1 2 3 4 5 6 7 8 9 10

**Figure 5.** (a) Time necessary for the participant to walk through the path configuration. (b) Number of collisions that happened to the participant during walking.**Table 2.** The number of responses (0–7) for each score (1–10) corresponds to the questionnaire answered by the participants after finishing the tests.

Scores questions	1	2	3	4	5	6	7	8	9	10
Q1	0	0	0	0	0	0	2	4	0	1
Q2	0	0	0	0	0	0	0	2	2	3
Q3	0	0	0	0	2	2	0	1	1	1
Q4	0	0	0	0	0	0	0	0	0	7
Q5	3	1	1	0	0	1	0	0	0	1
Q6	0	0	0	0	0	2	1	2	0	2

The tests were performed at Umeå University. A lecture hall was chosen with chairs and tables as obstacles to forming the paths. Outside the path, participants had five minutes of introductory time to get a short training on how the system works before they started with the tests. After completing the tests, participants were asked to answer a small questionnaire (Table 1). Their responses were collected anonymously.

Results

According to our preliminary results (Figure 5), the time necessary for the participants to walk through the path was shorter (the median (IQR) was 64s (32–81) versus 95s (51–134)) and slightly fewer collisions (the median (IQR) was 1 (0–2) versus 1 (0–4)) occurred once they used the presented assistive obstacle avoidance system compared to their baseline (i.e., using white cane only). The non-parametric Wilcoxon signed-rank tests were conducted on these data by comparing the time needed for walking through the paths and the number of collisions with versus without using the assistive obstacle avoidance system. The results show a significant difference in time (p -value < 0.05) but an insignificant difference in the number of collisions (p -value = 0.10).

Furthermore, the results of the questionnaire (Table 2) showed that most of the participants found the assistive system helpful (Q1) with a high potential for assisting in navigation (Q6), and many

participants felt trust in the system (Q3). According to their answers, the haptic feedback was intuitive (Q2), and they could easily distinguish vibrations between their right and left wrists (Q4). However, it was difficult to distinguish which vibrator vibrated on a unique wrist (Q5). This suggests that fewer vibrators are sufficient on each wrist, as the user is quite agnostic to which vibrator vibrates on a given wrist.

Moreover, experimental trials showed that the lithium battery used in the 'sensor module' could hold its charge for several days. The high energy consumption of the vibrators caused the 'haptic modules' to function for 3 h under full load before their batteries ran out of charge. However, given that our study showed that fewer vibrators can be used and that lithium batteries with higher capacity are available in the market, this duration can be easily extended to cover a full working day in realistic daily life situations.

Discussion

This paper presented a functional prototype of a haptic-based system to assist people with profound vision impairment in obstacle detection. The ability to detect multiple objects simultaneously with high precision and code them using multiple vibrations is an important feature of the developed system. The wireless communication between the 'sensor module' and the 'haptic modules' is also important because it allows the placement of these modules entirely independent of each other in the most appropriate and comfortable locations. For example, in this design, the 'sensor module' is mounted on the outer side of the belt which provides the best view to the laser sensors, and the 'haptic modules' are mounted in hand bands to code different distances and directions on both the right- and left-hand sides. It has been confirmed that the wrists and spine are the most sensitive locations for detecting vibrations across the body (Karuei et al., 2011). In addition, the direct contact between the mounted 'haptic modules' and the skin was shown to provide a good resolution for the perception of the vibrational signals by the user.

Another feature of the proposed system is that our 'haptic modules' code the distance to the object(s) using the PWM duty cycle. PWM is the predominant technique in control engineering when transferring variable power to a mechanical load (Sun, 2012). An experimental study on tactile stimulation on the abdomen using macro fiber composites compared the localization sensitivity of MFC actuators for several combinations of frequency and duty cycle (Phamduy et al., 2018). They found slightly better discrimination using 100 Hz with 50 percent duty cycle stimulation. As skin sensitivity is dependent on, among other factors, the waveform of the stimulus, PWM provides distinguishable power to the skin (Phamduy et al., 2018). In our proposed system, the actuator had an intrinsic frequency of 115 Hz at 3.7 V and the constant frequency of the PWM pulse was 89 Hz, whereas the duty cycle of the PWM signal increased as the object became closer. Figure 3 shows the relation between the power delivered by the haptic module to the user's skin as a function of distance for two 'n' settings. Accordingly, the system is capable of coding a certain range of distance by means of variations in the delivered power. Besides power variations, the changes in the harmonic frequency components generated by various pulse widths can also, to some extent, help the user distinguish the distances. As a result, the participants in the study stated that they found haptic feedback very intuitive (all participants answered question 2 with a score equal to or greater than 8 out of 10, see Table 2).

Our method is in contrast with the methods proposed by R ijezon et al. (2019) and Ranjbar et al. (2019), which present a simple sinusoidal vibration and increase the frequency of the vibration as the object gets closer. Devices such as Ultra Cane and Miniguide, also code the distance by varying the vibration frequency. Different sensitivity and frequency dependence of the human body's skin receptors to vibrotactile stimulations have been recognized and can be found in the literature (Cholewiak and Collins, 1991; Greenspan and Bolanowski, 1996). The skin tactile sensitivity is known to be dependent on many factors including the stimulus location on the body, its frequency, duration, and waveform, the contactor's area, and its surroundings as well as the subject's sex and age. Nevertheless, it has been shown that human skin is sensitive to vibrations up to 300 Hz, and its sensitivity is progressively reduced at higher frequencies (Mortimer et al., 2007). The limited frequency range of the human tactile perception may, therefore, impose a substantial limitation on systems that use solely frequency to code

distance. Our method, built on PWM, does not suffer from these limitations since it is based on keeping the fundamental frequency constant and varying the PWM duty cycle instead.

The employed laser sensors allow for a wide range of obstacle detection around the user up to 4 m and an approximately 65° diagonal FoV for each of the two lasers. However, the proposed system has some limitations. Among other factors, laser sensors are sensitive to extreme weather, which does not allow the system to be used outdoors in case of rare extreme weather conditions. In addition, a single type of laser sensor does not provide as high accuracy as offered using LiDARs or a combination of sensors such as in other studies (Chai & Lau, 2020; Liu et al., 2021; Maryono et al., 2021; Setiadi et al., 2020). On the other hand, higher accuracy comes with a higher cost and computational power, heavier equipment, and lower feasibility to be applicable in real life (Jain & Patel, 2023; Muhsin et al., 2024). For instance, a ground plane checking system has been developed by Chai and Lau (2020) using 1D LiDARs to detect ramps and staircases for visually impaired outdoor navigation. Liu et al. (2021) have developed a navigation system using a solid-state LiDAR camera with wireless bone-conduction earphones and a laptop placed in the blind person's backpack. They have achieved 43.6% accuracy with their developed system but as the system requires a lot of computational resources, it cannot be used for real-time applications. Another LiDAR-based navigation and object detection system for visually impaired people has been developed by Setiadi et al. (2020). In their system, they have used a neural network algorithm implemented in a mini-PC fed by information received by the camera and LiDAR sensors. The mini-PC and the sensors are mounted in a blind headset while a navigation belt guides the blind person. Additionally, they have used ultrasound, infrared, accelerometer, and RFID in their developed system which achieved a high object detection accuracy (87.5%). Maryono et al. (2021) have developed an object detection system using an IMU sensor, 2D LiDAR, Speaker, and power bank mounted on a hard hat that is worn by blind people. The detection ability of their system was reported to be 99.9% for objects in front of the users.

The proposed haptic feedback is based on an array of 2×3 vibration actuators. Nevertheless, it was found that using six vibrators was unnecessary because it was difficult for users to distinguish between the six vibrators on their wrists. This might be due to small distances between the actuators in a limited space on wrists. Spatial discrimination has been extensively studied on the wrist and other parts of the body (Cholewiak & Collins, 2003). It has been shown that placing actuators close together in space can cause a masking effect (Mortimer et al., 2007; Weisenberger, 1994). Our study suggests that an array of 2×2 actuators (with larger distances in between) on each wrist can be more discriminable. In addition, using fewer actuators leads to saving more charge in batteries, and the 'haptic modules' can run for a longer time before they need to be recharged. The results of this study indicated that 750 mAh Lithium batteries used in 'haptic modules' could run for 3 h under full load. Another way to increase the running time of 'haptic modules' is to use batteries with higher capacity. For instance, if we use 2000 mAh batteries the running time can be up to approximately 7.5 h. Furthermore, there was an input-output latency equal to or higher than 100 ms between the sensor and haptic modules, and some users could notice this amount of latency. Future work on optimizing the code for the microprocessors is suggested to decrease this latency to lower than 30 ms so that it will not be noticeable to the users. Studies in the field of acoustics (Saremi et al., 2023; Saremi & Stenfelt, 2011; Stone & Moore, 1999) have shown that the human brain cannot perceive latencies lower than about 30 ms.

To add a unique feature to the proposed system, we suggest allowing the user to set the ' n ' parameter in Eq. (1.a) based on their preferences and the landscape in which they were located. As Figure 3 illustrates, the choice of ' n ' determines how the system codes the distances. Accordingly, the user can vary the sensitivity of the system depending on the surrounding conditions to minimize unnecessary distractions. In particular, only those objects that are relevant to the user, that is, are close enough to the user and could potentially pose an imminent risk will be coded to avoid unnecessary distractions. As our analysis showed, at 1.5 m, the minimum height of the obstacle from the floor that the system codes is 3.7 cm. Smaller objects that lie on the floor closer than 1.5 m are supposed to be detected by the white cane. As stated earlier, it is important for the user to consider this assistive system as a complement to the white cane which can help them detect farther objects and also those nearby objects that lack a footprint on the floor. The proposed system, at least in its current form, should not be seen as a replacement for the white cane.

Finally, the path configuration used in our study was not a standardized mobility course. Further study on user evaluation of the presented assistive system is suggested using an established mobility course, such as that described by Roentgen et al. (2012b). They developed a path specifically for assessing the mobility performance of the user. Last but not least, our study was conducted in a small city which restricted the study to seven participants. Considering our small sample size, further clinical studies with a larger number of participants are essential to establish the statistical significance of the benefits of the proposed obstacle avoidance system for visually impaired people.

Conclusions

The proposed prototype is based on laser technology and provides a lightweight, low-cost, wearable, intuitive haptic feedback system based on the PWM technique. Furthermore, the communication between the sensor and haptic modules is fully wireless which provides enormous comfort for users when wearing the modules. The proposed system was tested on seven visually impaired individuals. The results showed that the participants could walk more quickly and with fewer collisions using the system compared with their baseline. According to a qualitative study performed on participants' answers to a questionnaire, the system was found to be reliable and a good complement to the white cane. These promising results suggest conducting statistical analyses of a larger sample size to comprehensively evaluate the efficiency of the proposed system and the user experience.

The presented study also suggests developing an Android mobile application to communicate with the user and be integrated with the 'sensor module' and 'haptic modules'. Utilizing an integrated mobile application can add even more value to the presented system in terms of assisting navigation and finding destinations in complex environments. Some recent studies support the beneficial aspects of using a mobile application by individuals with visual impairment for both indoor and outdoor navigation (Kahraman & Turhan, 2022; Nair et al., 2022).

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Author contributions statement

CRedit: **Ghazaleh Ghaffari**: Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **André Tagaro Andersson**: Data curation, Investigation, Methodology, Software, Writing – review & editing; **Per Hallberg**: Conceptualization, Data curation, Project administration, Resources, Writing – review & editing; **Amin Saremi**: Conceptualization, Data curation, Funding acquisition, Investigation, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Disclosure statement

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Data availability statement

The hardware design files as well as the developed software code for this study are available online on GitHub [https://github.com/andretagaro/Projekt_ogonfonden] for other researchers to further develop this project. The data that support the findings of this study are available from the corresponding author, [G.G.], upon reasonable request.

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